BIOLOGY BOARD QUESTION PAPER: MARCH 2022

Time: 3 Hrs. Max. Marks: 70

General Instructions:

1. Section A: Q.1 contains Ten MCQs of one mark each.
   i. For MCQs, you must write the correct answer with its alphabet (a/b/c/d).
   ii. Only first attempt will be evaluated.
   Q.2 contains Eight very short answer questions of one mark each.
2. Section B: Q.3 to Q.14 are short answer questions of two marks each.
3. Section C: Q.15 to Q.26 are short answer questions of three marks each.
4. Section D: Q.27 to Q.31 are long answer questions of four marks each.
5. Begin the answer of each section on a new page.

SECTION – A

Q.1 Select and write the correct answers: [10]

i. Number of meiotic/mitotic divisions during development of male gametophyte from MMC:
(A) One meiotic and two mitotic (B) Two meiotic only
(C) Two mitotic only (D) One mitotic and one meiotic

ii. During DNA replication, separated strands are prevented from recoiling by:
(A) single strand binding protein (B) reverse transcriptase
(C) endonuclease (D) polymerase

iii. Chromosomal aberration shown in diagram is:
(A) Deletion (B) Duplication (C) Inversion (D) Translocation

iv. ______ hormone causes efflux of K+ ions from guard cells and acts as antitranspirant.
(A) Gibberellins (B) Cytokinin (C) Ethylene (D) Abscissic acid

v. Test tube baby technique is:
(A) In-vitro fertilization (B) In-situ fertilization
(C) In-vivo fertilization (D) Artificial insemination

vi. Symptoms such as chest pain radiating to neck, jaw, left arm indicate:
(A) Malaria (B) Angina pectoris (C) Kidney failure (D) Typhoid

vii. Layer in close contact with CNS:
(A) Cranium (B) Dura mater (C) Arachnoid mater (D) Pia mater

viii. Cellular factors in innate immunity are provided by:
(A) phagocytes (B) antibody (C) T-lymphocyte (D) B-lymphocyte

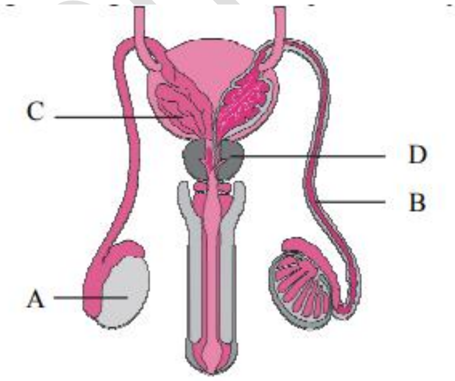
ix. Association representing brood parasitism:
(A) Hermit crab–sea anemone (B) Asian koel–Indian crow
(C) Algae–fungi (D) Buffalo–cattle egret

x. Annealing step of PCR operates at:
(A) 90–98°C (B) 40–60°C (C) 70–75°C (D) 100–120°C

Q.2 Answer the following questions: [8]
i. Name the part of gynoecium that determines pollen compatibility.
ii. Primary precursor of IAA in plants.
iii. Name the nitrogen-fixing cell in cyanobacteria.
iv. How many biodiversity hotspots identified worldwide?
v. Plant disease caused by Agrobacterium tumefaciens.
vi. Identify trophoblast cells in contact with embryonal knob during blastulation.
vii. Identify type of population growth curve from diagram.
viii. Meaning of Pioneer species.

SECTION – B

(Attempt any EIGHT) [16]
Q.3 Identify A, B, C, D in human reproductive system diagram.



Q.4 Identify chromosomal disorder due to non-disjunction of 21st chromosome and enlist symptoms.

Q.5 Write aims of Human Genome Project.

Q.6 Match parts of ovule (Column I) with seed parts (Column II):

	Column I		Column II
(a)	egg	(1)	testa
(b)	nucellus	(2)	tegmen
(c)	outer integument	(3)	perisperm
(d)	inner integument	(4)	embryo

Q.7 Enlist characteristics of Neanderthal Man.

Q.8 Define: (a) Gravitational water (b) Hygroscopic water (c) Combined water (d) Capillary water

Q.9 Give different properties of water.

Q.10 A person bleeds in a small accident but stops bleeding soon. Explain the physiological process.

Q.11 Match antibiotics with microbial sources:

Column I		Column II	
(a)	Chloromycetin	(1)	*Streptomyces griseus*
(b)	Erythromycin	(2)	*Streptomyces aurifaciens*
(c)	Streptomycin	(3)	*Streptomyces Venezuelae*
(d)	Terramycin	(4)	*Streptomyces erythreus*

Q.12 Abscissic acid = two identical substances isolated separately. Name them. Give chemical features.

Q.13 Effects of biotechnology on human health.

Q.14 Adaptations shown by desert animals.

SECTION – C

(Attempt any EIGHT) [24]

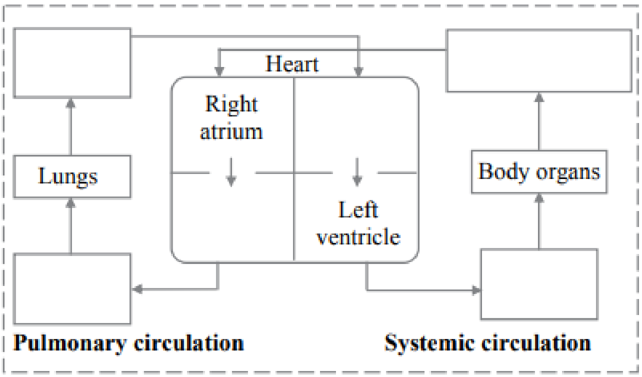
Q.15 Explain natural selection with example of industrial melanism.

Q.16 Describe physiological effects and applications of gaseous hormone in plants.

Q.17 What is ecological succession? Name various seral stages from pioneer to climax with example of hydrarch succession.

Q.18 Describe structure of root hair with neat labelled diagram.

Q.19 Complete flow diagram of double circulation.



Q.20 Distinguish between hyperthyroidism and hypothyroidism.

Q.21 Applications of DNA fingerprinting.

Q.22 Write a note on In-situ and Ex-situ conservation.

Q.23 Explain properties of nerve fibres.

Q.24 Give causative agent, mode of transmission and symptoms of typhoid.

Q.25 Match products with their microbial sources:

Products		Microbial Sources	
(a)	Vitamin B_2	(1)	*Rhizopus arrhizus*
(b)	Fumaric acid	(2)	*Candida lipolytica*
(c)	Vitamin B12	(3)	*Trichoderma Konigii*
(d)	Lipase	(4)	*Neurospora gossypii*
(e)	Cellulase	(5)	*Psuedomonas denitrificans*
(f)	Citric Acid	(6)	*Aspergillus niger*

Q.26 Explain any three examples of biopiracy.

SECTION – D

(Attempt any THREE) [12]

Q.27 Distinguish between artery and vein with neat labelled diagrams.

Q.28 State hormone and gland secreting it:

(a) Growth of thyroid gland

(b) Controls tubular absorption of water in kidney

(c) Stimulates liver & muscles for glycogenesis

(d) Development of immune system & T-lymphocyte maturation

Q.29 Describe outbreeding devices encouraging cross-pollination.

Q.30 Explain law of dominance and compare it with incomplete dominance & co-dominance.

Q.31 Describe hormonal control in phases of menstrual cycle.